

6. Cox process

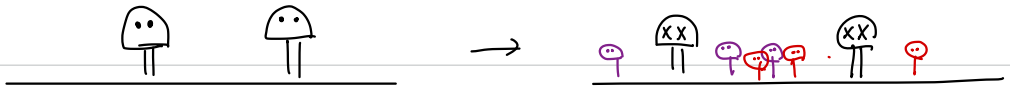
6.1 Definitions and basic properties

Cox process $\rightarrow$ doubly stochastic Poisson process.

S : state space Π : random countable subset.

Π is Cox process if the conditional distribution of Π given μ is Poisson process

Example. (in Section 5.5.)



$X + D_i$
 $\overline{\text{parents}} \quad \text{progeny}$
 poisson distribution

is conditionally
 Poisson process.

① disjoint.

$$\mathbb{P}\{N(A_1) = r_1, N(A_2) = r_2, \dots, N(A_n) = r_n \mid \mu\} = \prod_{k=1}^n \pi_{r_k}(\mu(A_k))$$

② disjoint.

Let random measure $\mu(A)$ be

$$\mu(A) = \int_A \underbrace{\Lambda(x)}_{\text{density}} dx$$

$\Lambda(x)$: real-valued random process on $S \rightarrow$ measurable.

so $\mu(A)$ well defined.

$$\begin{aligned} \mathbb{E}[N(A)] &= \mathbb{E}[\mathbb{E}[N(A) | \underbrace{\mu}_{\text{random measure}}]] = \mathbb{E} \int_A \Lambda(x) dx \\ &= \int_A \mathbb{E}[\Lambda(x)] dx. \end{aligned}$$

↙ (not too difficult to analysis)
 $\wedge$ many different non-Poisson random sets $\rightarrow$ Cox have flexible meaning

$$\begin{aligned} \mathbb{E}[N(A)^2] &= \mathbb{E}\{\mathbb{E}[N(A)^2 | \mu]\} \\ &= \mathbb{E}\{\mu(A) + \mu(A)^2\} \\ &= \mathbb{E}[\mu(A)] + [\mathbb{E}[\mu(A)]]^2 + \text{var}[\mu(A)] \end{aligned}$$

$$\text{var}[N(A)] = \text{var}[\mu(A)] + \mathbb{E}[\mu(A)].$$

IF $\mu(A)$ degenerate random variable

$\rightarrow \text{var}[N(A)] = \mathbb{E}[\mu(A)]$ same with (poisson distribution)

$N(A)$ has greater variance than Poisson.

6.2 Cox process in ecology

non-uniform rate function $\lambda(x)$.

$\lambda(x)$ be considered known $\rightarrow$ non-homogeneous Poisson process

But in ecology example, $\rightarrow \lambda(x)$ is not known (random factor parents.)

X parents x daughters

X parents x daughters
number of daughters $\rightarrow$ poisson distribution mean is $\phi(X)$.

$$\frac{X + D_i}{\sigma} \rightarrow \text{probability } g(X, x - X)$$

Let $\phi(x)$ is constant & $g(x, x-x) = g(x-x)$.

$$\Lambda(x) = \phi \sum g(x - X).$$

1st generation X 2nd generation X' 3rd generation X''

rate ftn.

 $\wedge$

✓

$$\Lambda'(x) = \phi \sum g(x - x_j).$$

$$E[\Lambda'(x) | \Lambda] = \int_{\mathbb{R}^2} \phi g(x-y) \Lambda(y) dy$$

$$\mathbb{E}[\Lambda'(x)] = \int_{\mathbb{R}^2} \phi g(x-y) \mathbb{E}[\Lambda(y)] dy$$

$$\text{If } \mathbb{E}[\Lambda(x)] = \lambda \rightarrow \mathbb{E}[\Lambda'(x)] = \lambda \phi.$$

$$\boxed{\text{Let } \phi = 1}$$

(8.12)

$$\Sigma_1 = \sum_{x \in \Pi} f_1(x) \quad \Sigma_2 = \sum_{x \in \Pi} f_2(x)$$

$$\mathbb{E}(\Sigma_1 \Sigma_2) = \int f_1 d\mu \int f_2 d\mu + \int f_1 f_2 d\mu.$$

$$\begin{aligned} \frac{1}{\lambda} \mathbb{E}(\Lambda'(x) \Lambda'(y) | \Lambda) &= \int_{\mathbb{R}^2} g(x-\xi) \Lambda(\xi) d\xi \int_{\mathbb{R}^2} g(y-\eta) \Lambda(\eta) d\eta \\ &\quad + \int_{\mathbb{R}^2} g(x-\xi) g(y-\xi) \Lambda(\xi) d\xi \end{aligned}$$

$$\begin{aligned} \mathbb{E}(\Lambda'(x) \Lambda'(y)) &= \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} g(x-\xi) g(y-\eta) \mathbb{E}(\Lambda(\xi) \Lambda(\eta)) d\xi d\eta \\ &\quad + \lambda \int_{\mathbb{R}^2} g(x-\xi) g(y-\xi) d\xi \end{aligned}$$

Λ is second-order stationary. $\because$ mean constant

$$\mathbb{E}[\Lambda(x) \Lambda(y)] = \lambda^2 + \sigma(y-x). \quad \left\{ \begin{array}{l} \text{covariance only depends on} \\ \text{difference} \end{array} \right.$$

$$\mathbb{E}(\Lambda'(x) \Lambda'(y)) = \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} g(x-\xi) g(y-\eta) (\lambda^2 + \sigma(\eta-\xi)) d\xi d\eta \\ + \lambda \int_{\mathbb{R}^2} g(x-\xi) g(y-\xi) d\xi$$

$$\sigma'(y-x) \\ = \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} g(x-\xi) g(y-\eta) \sigma(\eta-\xi) d\xi d\eta + \lambda \int_{\mathbb{R}^2} g(x-\xi) g(y-\xi) d\xi \\ y-x = t \quad a = y-x+\xi-\eta \quad b = y-\eta \quad k = y-\xi$$

$$\sigma'(t) = \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \sigma(t-a) g(b-a) g(b) db da + \lambda \int_{\mathbb{R}^2} g(k) g(k-t) dk \\ = \int_{\mathbb{R}^2} \sigma(t-a) \gamma(a) da + \lambda \gamma(t)$$

distribution g has finite variance $\sigma(0) = \text{var}[\Lambda(x)]$

without limit as the generation go by.

6.3 The Borel-Tanner distribution.



Population which reproduces from generation to generation in a Poisson way.

family size distribution $P(\phi)$ $\phi \leq 1 \rightarrow$ finite number of descendants.

Consider Galton-Watson Process. - single ancestor.

$$Z_0 = 1 \text{ (ancestor)}$$

$$P(Z_1 = k) = p_k$$

$$f(z) = \sum_{k=0}^{\infty} p_k z^k.$$

$$\mathbb{E}(z^N) = \sum_{k=0}^{\infty} p_k z^k = f(z). \quad \text{← } z, 1 \text{에 대한 generating. f.n.}$$

$$f_{(n+1)}(s) = f_{(n)}[f(s)] \quad \text{↻} \quad f_0(s) = s \text{ 라 하자.}$$

$$f'(1) = \sum_{x=0}^{\infty} x p(x) = \text{mean family size.} \leq 1$$

→ only a finite number of non-empty generations.

* The generating function on Z_n is the n th iterate $f_n(s)$.

pf.

Let $f_n(s)$ designate the generating function of Z_n .

under the condition $Z_n = k$ each k th daughter independent generation $f_n \rightarrow [f(s)]^k$.

$$f_{(n+1)}(s) = \sum_{k=0}^{\infty} P(Z_n = k) [f(s)]^k = f_n(f(s))$$

$$P(Z_n \rightarrow 0) = P(Z_n = 0 \text{ for some } n) \rightarrow \{Z_n = 0\} \rightarrow Z_{n+1} = 0.$$

$$= P[(Z_1 = 0) \cup (Z_2 = 0) \cup \dots] = \lim_{n \rightarrow \infty} P(Z_n = 0)$$

$$= \lim_{n \rightarrow \infty} P(Z_n = 0) = \lim_{n \rightarrow \infty} f_n(0) = q$$

$q = f(q)$ → find the fixed point!

$$m = f'(1)$$

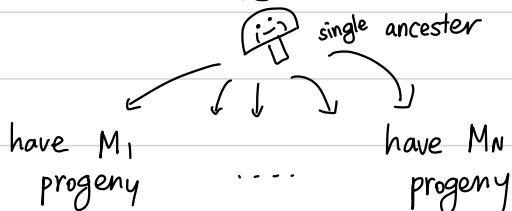
$$f'(s) < f'(1)$$

$$m \leq 1 \rightarrow f'(s) < 1$$

$$f(s) < s \text{ for } 0 \leq s < 1$$

$$\Rightarrow q = 1$$

If Ancestor have N children.



$$M = 1 + M_1 + \dots + M_N$$

Let $G(z) = \mathbb{E} [\mathbb{E}(z^M | N)]$.

generating function of M when

Ancestor have N children.

$$\begin{aligned} \mathbb{E}(z^M | N) &= z \cdot \sum p(z_1 = M_1, \dots, z_N = M_N) z^{M-1} \\ &= z \sum_{M_1} p(z_1 = M_1) z^{M_1} \dots \sum_{M_N} p(z_N = M_N) z^{M_N} \\ &= z G(z)^N. \end{aligned}$$

$$G(z) = \mathbb{E}(z G(z)^N) = z F[G(z)].$$

How Can we solve this? $\rightarrow$ Whittaker and Watson.

Using Lagrange's expansion.

$$\xi = a + t \phi(\xi) \rightarrow f(\xi) = f(a) + \sum_{n=1}^{\infty} \frac{t^n}{n!} \frac{d^{n-1}}{da^{n-1}} [f'(a) [\phi(a)]^n]$$

So

$$G(z) = \sum_{n=1}^{\infty} \frac{z^n}{n!} \frac{d^{n-1}}{dx^{n-1}} [F(x)^n]_{x=0}$$

$$f(G(z)) = f(0) + \sum_{n=1}^{\infty} \frac{z^n}{n!} \frac{d^{n-1}}{dx^{n-1}} [F(x)^n f'(x)]_{x=0}$$

$$G(z)^r = \sum_{n=1}^{\infty} \frac{z^n}{n!} \frac{d^{n-1}}{dx^{n-1}} [F(x)^n \cdot r x^{r-1}]_{x=0}$$

$\hookrightarrow$ Generating function of total progeny $M^{(r)}$ r sister ancestors

$$\mathbb{P}[M^r = n] = \frac{r}{n!} \frac{d^{n-1}}{dx^{n-1}} [F(x)^n \cdot r x^{r-1}]_{x=0}$$

$F(x)^n \rightarrow$ generating function of the total size S_n of n different families

$$\frac{d^{n-1}}{dx^{n-1}} [F(x)^n x^{r-1}]_{x=0} \rightarrow F(0) \quad n-r \triangleq \text{계수 pick up.}$$

$$IP[M^r = n] = \frac{r}{n} IP[S_n = n-r]$$

In particular,

$$IP[M = n] = \frac{1}{n} IP[S_n = n-1]$$

family size distribution is $P(\phi) \rightarrow S_n$ has distribution $P(n\phi)$

$$IP[M^{(r)} = n] = \frac{r}{n} \pi_{n-r}(n\phi)$$

$$IP[M = n] = \frac{1}{n} \pi_{n-1}(n\phi)$$

Borel
-Tanner
distribution.

$$\sum_{n=r}^{\infty} \frac{\pi_{n-r}(nt)}{n} = \frac{1}{r} \quad \text{probability sum is 1}$$

$$\sum_{n=1}^{\infty} \frac{n^{n-1} t^{n-1} e^{-nt}}{n} = 1$$

$$\sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} (te^{-t})^n = t$$

$$\psi(x) = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} x^n$$

$|x| \leq e^{-1}$
 $\rightarrow$ convergent.

$$te^{-t} = s \rightarrow t = \psi(s)$$

6.4 Cox processes and renewal processes

state space.

Let $S = \mathbb{R}$.

Specifically λ be a positive fn. s.t.

$$\int_{-\infty}^{\infty} \lambda(x) dx = \int_0^{\infty} \lambda(x) dx = \infty.$$

$\Psi: \mathbb{R} \rightarrow \mathbb{R}$ unique right-continuous function

$$\int_0^{\Psi(y)} \lambda(x) dx = y$$

Π : Poisson process of constant 1 on $\mathbb{R}$. (Section 4.5)

The points $\Psi(X)$ ($X \in \Pi$) form a Poisson process with rate $\lambda(x)$.

To construct Cox process.

$\lambda \rightarrow$ random process Λ independent of Π

$\{\Psi(X); X \in \Pi\} \rightarrow$ Cox process.

[Use Ψ to transform the homogeneous Poisson process Π .
Samples the process Ψ at Poisson instants $X \in \Pi$ ★

* Poisson sampling loses no information about joint distributions.

* No means obvious (Kingman, 1964)

Renewal distribution $\rightarrow$ Cox process

random subsets

$0 < X_1 < X_2 < \dots$ s.t. $X_1, X_2 - X_1, X_3 - X_2, \dots$ are independent.

but have common 'lifetime' distribution

$X_1, X_2, \dots \rightarrow$ Cox processes if lifetime distribution have special form.

probability density $\lambda p(t)$.

$$p(t) = \mathbb{P} \{ \xi(t) = 0 \mid \xi(0) = 0 \}.$$

[Markov process. with $p(0+) = 1$.

$$\lambda(t) = \lambda \quad \text{if } \xi(t) = 0$$

$$= 0 \quad \text{otherwise.}$$

'most' interesting random countable sets are, if not Poisson, at least members of the Cox family.